

JUNIOR CERTIFICATE

MATHEMATICS

Form 2

Study Notes

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TOPIC 1 ARITHMETIC SEQUENCE

- A sequence in which any two consecutive terms differ by the same number, i.e. a constant, is called an arithmetic sequence.
- The constant number by which the consecutive terms differ is called the common difference.
- An arithmetic sequence is generally expressed in the form:
 $a, a + d, a + 2d, a + 3d, \dots, a + (n - 1)d$, where a is the first term and d is the common difference.
- In general, the arithmetic sequence is given by the formula:
nth term = $a + (n - 1)d$
where: a = first term ,
 n = number of terms
 d = common difference.
- In an arithmetic sequence, the difference between two succeeding terms is always the same.

Example

Find the 17th term in the arithmetic sequence 12, 9, 6,

Solution

Given $a = 12$ and $d = -3$.

Using nth term = $a + (n - 1)d$, you obtain

$$17 \text{ th term} = 12 + (17 - 1) \times (-3)$$

$$= 12 - 48$$

$$= -36.$$

The 17th term is -36

Example

In the sequence $-2, 2, 6, 10, \dots$, what is the 16th term ?

Solution

First term, $a = -2$, common difference, $d = 2 - (-2) = 4$,

n th term $= a + (n - 1) d$

$$\begin{aligned}\therefore 16\text{th term} &= -2 + (16 - 1) \times 4 \\ &= 58.\end{aligned}$$

Example

The seventh term of an arithmetic sequence is 80. If the first term is 98, what is the 4th term?

Solution

Using $T_n = a + (n - 1) d$

$n = 7$, $T = 80$ and $a = 98$

$$80 = 98 + (7 - 1)d$$

$$-6d = 98 - 80$$

$$\therefore d = -3$$

$$4\text{th term} = 98 + -3(4 - 1)$$

$$= 98 + 3(-3)$$

$$= 98 - 9$$

$$= 89$$

Activity

1. Are the following sequences arithmetic? If so, state the common difference and the seventh term.

(i) 27, 29, 31, 33, ... (ii) 1, 2, 3, 5, 8, ... (iii) 2, 4, 8, 16, (iv) 3, 7, 11, 15, ...

2. The first term of an arithmetic sequence is -8 and the common difference is 3. Find the seventh term of the sequence. The last term of the sequence is 100. How many terms are there in the sequence?

3. The first term of an arithmetic sequence is 12, the seventh term is 36 and the last term is 144. Find the common difference and the number of terms in the sequence.

4. The terms -8, P, 12, 22 are in a sequence. Calculate the value of P.

TOPIC 2 QUADRATIC EXPRESSIONS AND EQUATIONS

Factorization of quadratic expression

Example

Factorize $3m + 3n$

$= 3(m + n)$ (the common term is 3 so we put it outside the bracket)

Factorization by grouping

When the terms of an expression do not have a common factor, the terms are put into pairs to find a common factor. This method is known as factorization by grouping.

Example

Factorize:

a. $3ab + 2b + 3ca + 2c$

Solution

$3ab + 2b + 3ca + 2c$

$= b(3a + 2) + c(3a + 2)$

$= (3a + 2)(b + c)$

b. $ab + bx - a - x$

Solution

$$\begin{aligned}
 & ab + bx - a - x \\
 &= b(a + x) - 1(a + x) \\
 &= (a + x)(b - 1)
 \end{aligned}$$

Factorising a Quadratic expression

- To factorize the expression, we look for two numbers such that their product is ac and their sum is b . a , b are the coefficient of x while c is the constant.
- The middle term is the sum of two terms.

Example

$(x + 2)(x + 3)$ is multiplied out as:

$$\begin{aligned}
 & (x + 2)(x + 3) \\
 &= x(x + 3) + 2(x + 3) \\
 &= (x)(x) + 3x + 2x + (2)(3) \\
 &= x^2 + 5x + 6.
 \end{aligned}$$

Example

Factorise $x^2 + 4x + 4$

Find two number such that their product is 4

Their sum is 4 where 4 is the coefficient of x

The numbers are 2 and 2 Rewrite $4x$ as $2x + 2x$ thus,

$$\begin{aligned}
 & x^2 + 2x + 2x + 4 \\
 &= x(x + 2) + 2(x + 2) \\
 &= (x + 2)(x + 2)
 \end{aligned}$$

Example

Factorise : $8x^2 + 10x + 3$

Solution

$$\begin{aligned}
 & 8x^2 + 10x + 3 \\
 &= 8x^2 + 4x + 6x + 3 \\
 &= 4x(2x + 1) + 3(2x + 1) \\
 &= (4x + 3)(2x + 1)
 \end{aligned}$$

Example

Factorize : $6x^2 - 13x + 6$

Solution

$$\begin{aligned} & 6x^2 - 13x + 6 \\ &= 6x^2 - 4x - 9x + 6 \\ &= 2x(3x - 2) - 3(3x - 2) \\ &= (2x - 3)(3x - 2) \end{aligned}$$

Difference of two squares

The third simplest quadratic is made up of the difference of squares. We know that:

$$(a + b)(a - b) = a^2 - b^2.$$

This is true for any values of a and b , and more importantly since it is an equality, we can also write: $a^2 - b^2 = (a + b)(a - b)$.

Example

Factorise $9x^2 - 25$

Step 1 : Examine the quadratic expression. We see that the quadratic is a difference of Squares because: $(3x)^2 = 9x^2$ and $5^2 = 25$.

Step 2 : Write the quadratic as the difference of squares

$$9x^2 - 25 = (3x)^2 - 5^2$$

Step 3 : Write the factors

$$(3x)^2 - 5^2 = (3x - 5)(3x + 5)$$

Quadratic Equations

A quadratic equation is any expression of the form $ax^2 + bx + c$, $a \neq 0$.

Example

Solve: $x^2 + 3x - 54 = 0$

Solution

Factorize the left hand side

$$x^2 + 9x - 6x - 54 = 0$$

$$(x - 6)(x + 9) = 0$$

$$x - 6 = 0 \text{ or } x + 9 = 0$$

Hence $x = 6$ or -9

Word problems involving quadratic equations

Example

A rectangular room is 4 m longer than it is wide. If its area is 12 find its dimensions.

Solution

Let the width be x m .its length is then $(x + 4)$ m.

The area of the room is $x(x+4)$ Therefore $x(x + 4) = 12$

$$x^2 + 4x = 12$$

$$x^2 + 4x - 12 = 0$$

$$x^2 + 6x - 2x + 2 = 0$$

$$x(x+6) - 2(x+6) = 0$$

$$(x-2)(x+6) = 0$$

$$x = 2, x = -6$$

-6 is being ignored because length cannot be negative

The length of the room is $x + 4 = 2 + 4$

$$= 6\text{m}$$

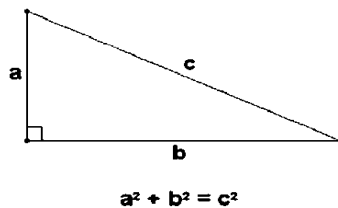
Activity

1. When x is added to its square, the answer is 12. Find the two possible values of x .
2. When x is subtracted from its square, the answer is 20. Find the possible values of x .
3. Simone thinks of a number. She squares the number and adds twice the original number.
4. A rectangle has a length of $(x + 5)$ cm and a width of x cm. Write down an expression for the area of the rectangle. , find the value of x .
5. The product of two consecutive odd numbers is 143. Find two positive numbers and two negative numbers that satisfy the condition.
6. The length of a picture is 10 cm more than the width. The area is 1200 cm. Find the dimensions of the picture.
7. Two integers differ by 6. The sum of the squares of these integers is 116. Find the two integers.

TOPIC 3 PYTHAGORAS THEOREM

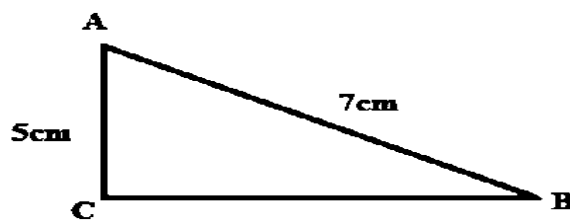
- Pythagoras theorem states that for a right-angled triangle, the square of the hypotenuse is equal to the sum of the square of the two shorter sides.

Consider the triangle below:



Example

Given that the triangle ABC, angle ACB = 90°, AB = 7cm and AC = 5 cm, calculate the length of BC giving your answer correct to one decimal place.



Solution

Using Pythagoras theorem

$$a^2 + b^2 = c^2$$

$$a^2 + 5^2 = 7^2$$

$$a^2 = 49 - 25$$

$$= 24$$

$$a = \sqrt{24}$$

$$= 4.9\text{cm}$$

Example

In a right angle triangle, has two shorter sides 6 cm and 8 cm. Find the length of the other side.

Solution

$$c^2 = a^2 + b^2$$

$$c = 6^2 + 8^2$$

$$c = 36 + 64$$

$$c = 100 = \sqrt{100} = 10\text{cm}$$

Example

In a triangle ABC, AB=7cm, AC= 3 cm and angle ACB=90°. Calculate the length of BC giving your answer correct to 2 significant figures.

Solution

$$AB^2 + BC^2 = AC^2$$

$$7^2 + BC^2 = 3^2$$

$$BC^2 = 9 - 49$$

$$BC^2 = -40$$

$$= 2\sqrt{10}$$

$$= 2 \times 3.162$$

$$= 6.324 \approx 6.3 \text{ (2 s. f.)}$$

Example

In a triangle XYZ, X = 3 cm, Y = 4 cm and X = 5 cm. Show that triangle XYZ is right angled.

Solution

$$XY^2 + YZ^2 = XZ^2$$

$$= 3^2 + 4^2$$

$$= 9 + 16$$

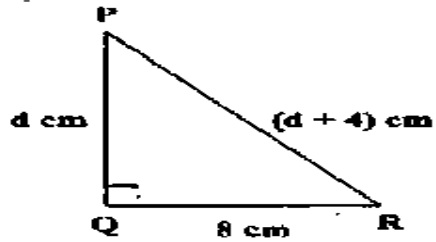
$$= 25$$

$$= 5^2$$

∴ The triangle XYZ is right angled.

Example

Figure below is a right angled triangle PQR in which $PQ = d$ cm, $QR = 8$ cm and $PR = (d + 4)$ cm.



Calculate the value of d .

Solution

$$(d + 4)^2 = d^2 + 8^2$$

$$d^2 + 8d + 16 = d^2 + 64$$

$$8d = 48$$

$$d = 6$$

TOPIC 4 CONGRUENCY IN TRIANGLES

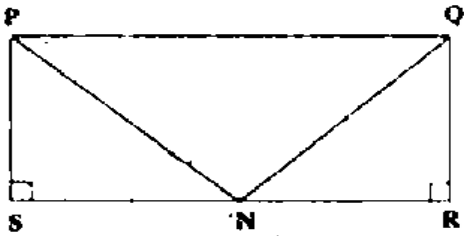
- Congruent triangles are those triangles that have corresponding sides and corresponding angle equal.
- The symbol for congruent triangles is $\equiv$ or $\cong$

Proving congruency in triangles

Theorem 1: Side angle side (SAS)

Example

In Figure below PQRS is a rectangle.



If $SN = RN$, show that triangle PSN is congruent to triangle QRN.

Solution

$SN = RN$ (given)

$PS = QR$ (PQRS is a rectangle)

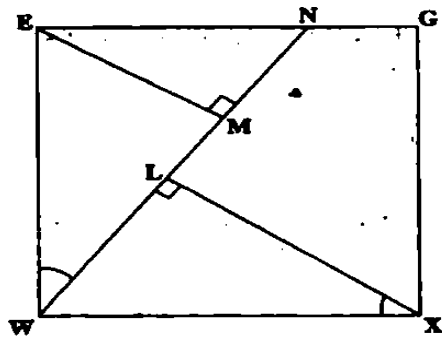
$\angle PSN = \angle QRN = 90^\circ$ (PQRS is a rectangle)

$\therefore$ Triangle PSN $\equiv$ Triangle QRN (S. A. S)

Theorem 2 : Angle, angle, side (AAS)

In Figure 3 N lies on a side of the square EGXW. EM and XL are perpendicular to WN. Angle

$\angle EWM = \angle WXL$.



Show that triangle WME and XLW are congruent.

Solution

In $\triangle WME$ and $\triangle XLW$,

$$\angle WME = \angle XLW = 90^\circ$$

(EM and XL are perpendicular to WN)

$$\angle EWM = \angle WXL \text{ (given angle)}$$

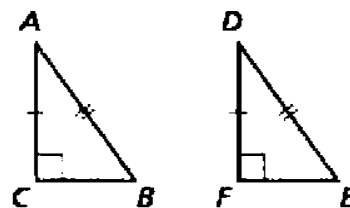
$WE = XW$ (sides of a square are equal)

$$\therefore \triangle WME \cong \triangle XLW \text{ (A.A.S)}$$

Theorem 3: Right angle hypotenuse side (RHS)

If the hypotenuse and a leg of a right triangle are congruent to the hypotenuse and a leg of a second right triangle, then the two triangles are congruent.

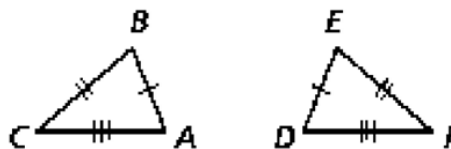
If $\overline{AB} \cong \overline{DE}$, $\overline{AC} \cong \overline{DF}$, and $m\angle C = m\angle F = 90^\circ$, then $\triangle ABC \cong \triangle DEF$.



Theorem 4: Side, side, side (SSS)

If three sides of one triangle are congruent to three sides of a second triangle, then the two triangles are congruent.

If $\overline{AB} \cong \overline{DE}$, $\overline{BC} \cong \overline{EF}$, and $\overline{AC} \cong \overline{DF}$, then $\triangle ABC \cong \triangle DEF$.



TOPIC 5 PLANE SHAPES

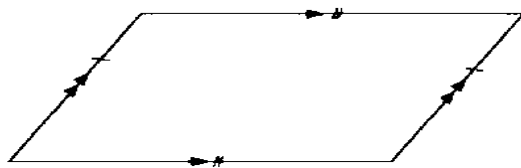
Quadrilaterals

- A quadrilateral is a four –sided plane figure.
- The interior angles of a quadrilateral add up to 360° .
- Quadrilaterals are also classified in terms of sides and angles.

Properties of quadrilaterals

Properties of Parallelograms

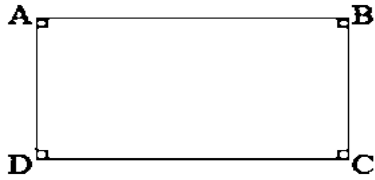
A parallelogram is a quadrilateral in which opposite sides are equal and parallel



- The parallel sides are parallel by definition.
- The opposite sides are congruent.
- The opposite angles are congruent.
- The diagonals bisect each other.
- Any pair of consecutive angles are supplementary.

Properties of Rectangles

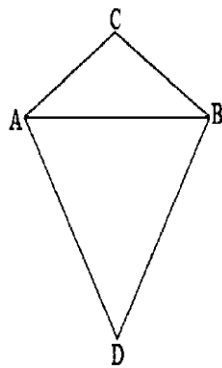
- A rectangle is a quadrilateral in which opposite sides are equal and parallel, and all angles are right angles



- All the properties of a parallelogram apply by definition.
- All angles are right angles.
- The diagonals are congruent.

Properties of a kite

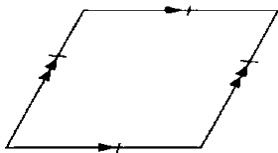
- A kite is a quadrilateral with pairs of adjacent equal sides.



- Two disjoint pairs of consecutive sides are congruent by definition.
- The diagonals are perpendicular.
- One diagonal is the perpendicular bisector of the other.
- One of the diagonals bisects a pair of opposite angles.
- One pair of opposite angles are congruent.

Properties of a rhombus

- A rhombus (plural: rhombi) is a parallelogram with all the four sides equal in length

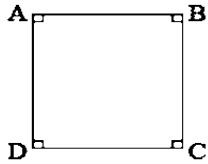


- All the properties of a parallelogram apply by definition.
- Two consecutive sides are congruent by definition.
- All sides are congruent.
- The diagonals bisect the angles.

- The diagonals are perpendicular bisectors of each other.
- The diagonals divide the rhombus into four congruent right triangles.

Properties of a square

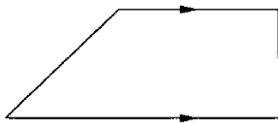
A square is a rectangle in which all sides have the same length.



- All the properties of a rectangle apply by definition.
- All the properties of a rhombus apply by definition.
- The diagonals form four isosceles right triangles.

Properties of trapezium

- A trapezium (plural: trapezia) is a quadrilateral which has at least one pair of opposite sides parallel.
- If the remaining two sides are equal, the trapezium is said to be isosceles.



Theorems in quadrilaterals

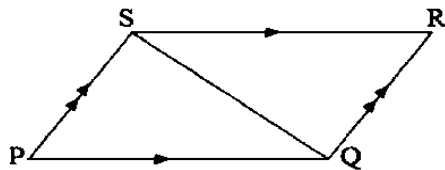
A theorem is a statement that can be proven to be true by way of experiment.

Theorem 1 : A diagonal of a parallelogram divides it into two congruent triangles (two equal parts).

Theorem 2: In a parallelogram, opposite sides are equal.

Given a parallelogram PQRS, prove that $PQ = SR$ and $PS = QR$.

Proof: Draw parallelogram PQRS



Join SQ.

In Δ s PQS and QSR

$\angle PQS = \angle QSR$ (alternate angles, $PQ \parallel SR$)

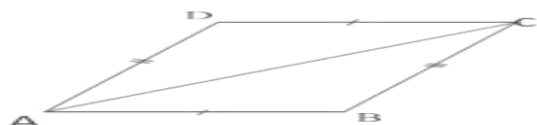
$\angle PSQ = \angle SQR$ (alternate angles, $PS \parallel QR$)

$SQ = QR$ (common line)

$\therefore \triangle PQS$ and $\triangle QSR$ are congruent (AAS or ASA).

Hence, $PQ = SR$ and $PS = QR$.

Theorem 3: If each pair of opposite sides of a quadrilateral is equal, then it is a parallelogram.



Let sides AB and CD of the quadrilateral $ABCD$ be equal and also $AD = BC$ (see Fig. above). Draw diagonal AC .

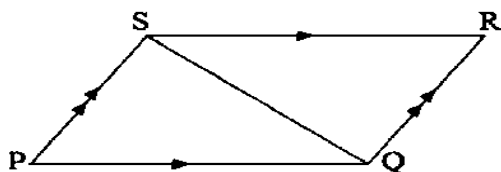
Clearly, $\triangle ABC \cong \triangle CDA$ (already proved)

So, $\angle BAC = \angle DCA$ (alternate angles) and $\angle BCA = \angle DAC$ (alternate angles)

Therefore, $ABCD$ is a parallelogram

Theorem 4 : In a parallelogram, opposite angles are equal.

Using the parallelogram in figure below , prove that $\angle SPQ = \angle SRQ$ and $\angle PSR = \angle PQR$.



Proof: $\angle SPQ + \angle PSR = 180^\circ$ (supplementary angles, $PQ \parallel SR$)

$\angle SPQ + \angle PQR = 180^\circ$ (supplementary angles, $PS \parallel QR$)

Thus, $\angle SPQ + \angle PSR = \angle SPQ + \angle PQR = 180^\circ$

hence, $\angle PSR = \angle PQR$

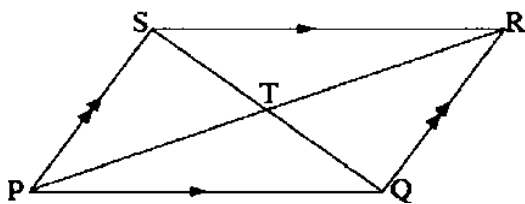
Similarly, $\angle SPQ = \angle SRQ$

Diagonals bisect each other.

Theorem 5 : If in a quadrilateral, each pair of opposite angles is equal, then it is a parallelogram.

Theorem 6 The diagonals of a parallelogram bisect each other.

Given a parallelogram $PQRS$ with diagonals SQ and PR intersecting at point T , prove that $ST = TQ$ and $PT = TR$.



Proof: In Δ s STR and PTQ,

$\angle RST = \angle TQP$ (alternate angles, $SR \parallel PQ$)

$\angle TRS = \angle TPQ$ (alternate angles, $SR \parallel PQ$)

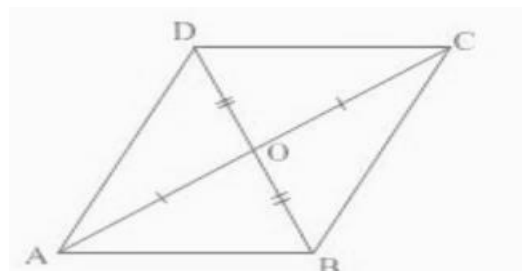
$SR = PQ$ (opposite sides of a parallelogram)

$\therefore$ STR and Δ PTQ are congruent (ASA)

Hence, $ST = TQ$ and $PT = TR$.

Theorem 7 : If the diagonals of a quadrilateral bisect each other, then it is a parallelogram.

Using the figure below, it is given that $OA = OC$ and $OB = OD$.



So, $\Delta AOB \cong \Delta COD$

Therefore, $\angle ABO = \angle CDO$ (alternate angles)

From this, we get $AB \parallel CD$

Similarly, $BC \parallel AD$

Therefore ABCD is a parallelogram

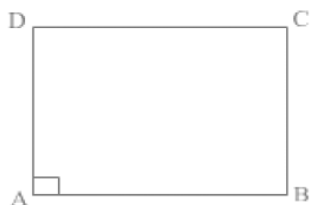
Example

Show that each angle of a rectangle is a right angle.

Solution

Let us recall what a rectangle is.

A rectangle is a parallelogram in which one angle is a right angle.



Let ABCD be a rectangle in which $A = 90^\circ$.

To show that $B = C = D = 90^\circ$

$AD \parallel BC$ and AB is a transversal

So, $A + B = 180^\circ$ (Interior angles on the same side of the transversal)

But, $A = 90^\circ$

So, $B = 180^\circ - A = 180^\circ - 90^\circ = 90^\circ$

$C = A$ and $D = B$ (Opposite angles of the parallelogram)

So, $C = 90^\circ$ and $D = 90^\circ$.

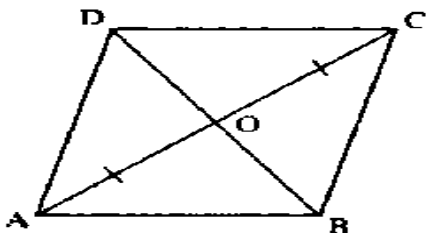
Therefore, each of the angles of a rectangle is a right angle.

Example

Show that the diagonals of a rhombus are perpendicular to each other.

Solution

Consider the rhombus ABCD below.



$AB = BC = CD = DA$ (corresponding sides of congruent triangles)

In $\triangle AOD$ and $\triangle COD$,

$OA = OC$ (Diagonals of a parallelogram bisect each other)

$OD = OD$ (Common) $AD = CD$

Therefore, $\triangle AOD \cong \triangle COD$ (SSS congruence rule)

This gives, $\angle AOD = \angle COD$

But, $\angle AOD + \angle COD = 180^\circ$ (Linear pair) So, $2 \angle AOD = 180^\circ$ or, $\angle AOD = 90^\circ$

So, the diagonals of a rhombus are perpendicular to each other.

Example

ABC is an isosceles triangle in which $AB = AC$. AD bisects exterior angle PAC and $CD \parallel AB$ (see Fig. that follows).

Show that

(i) $\angle DAC = \angle BCA$ and (ii) ABCD is a parallelogram.

Solution

(i) $\triangle ABC$ is isosceles in which $AB = AC$ (Given)

So, $\angle ABC = \angle ACB$ (Angles opposite to equal sides)

Also, $\angle PAC = \angle ABC + \angle ACB$

(Exterior angle of a triangle)

or, $\angle PAC = 2 \angle ACB$ ----- (1)

Now, AD bisects $\angle PAC$.----- (2)

So, $\angle PAC = 2 \angle DAC$

Therefore, $2 \angle DAC = 2 \angle ACB$ [From (1) and (2)] or, $\angle DAC = \angle ACB$

(ii) Now, these equal angles form a pair of alternate angles when line segments BC and AD are intersected by a transversal AC . So, $BC \parallel AD$ Also, $BA \parallel CD$

(Given)

Now, both pairs of opposite sides of quadrilateral $ABCD$ are parallel. So, $ABCD$ is a parallelogram.

Example

Two parallel lines l and m are intersected by a transversal. Show that the quadrilateral formed by the bisectors of interior angles is a rectangle.

Solution

It is given that $PS \parallel QR$ and transversal p intersects them at points A and C respectively.

The bisectors of $\angle PAC$ and $\angle ACQ$ intersect at B and bisectors of $\angle ACR$ and $\angle SAC$ intersect at D .

We are to show that quadrilateral $ABCD$ is a rectangle.

Now, $\angle PAC = \angle ACR$ (Alternate angles as $l \parallel m$ and p is a transversal)

Theorem 8: A quadrilateral is a parallelogram if a pair of opposite sides is equal and parallel.

Look at Fig 8.17 in which $AB = CD$ and $AB \parallel CD$. Let us draw a diagonal AC . You can show that $\triangle ABC \cong \triangle CDA$ by SAS congruence rule.

So, $BC \parallel AD$ (corresponding sides in a parallelogram)

And AB is also parallel to DC (same reason as above)

Therefore $ABCD$ is a parallelogram

TOPIC 6 MENSURATION

- A solid is a figure that has length, breadth (or width) and thickness.
- In other words, a solid is a figure that is not a plane (or flat) and therefore occupies space, e.g. a cube or pyramid.
- The three measures of a solid namely length, breadth and thickness are called its dimensions.

Naming solids

- Solids with many plane faces are called polyhedra.
- The singular is polyhedron, which means ‘having many faces.’
- The following are names and descriptions of some common solids.

Cuboid

- It is a solid bounded by three pairs of identical faces which are all rectangles.
- Sometimes, a cuboid is referred to as a ‘rectangular box’ or a ‘rectangular block.’

Cube

- It is a solid bounded by six identical faces which are all squares.
- A cube is a special type of cuboid, and it may sometimes be called a ‘square box’.

Pyramid

- It is a solid figure with triangular slanting faces which meet at one point, called an apex or vertex, above a polygonal base.
- A pyramid is always named after the shape of its base.

- Thus, there are triangular-based pyramids (tetrahedra), rectangular-based pyramids, pentagonal-based pyramids, and so on.
- If the pyramid has its vertex vertically above the centre of the base, it is called a right pyramid.

Tetrahedron

- This is a solid figure with four faces, all triangles.
- Therefore, it is a pyramid with a triangular base.

Prism

- It is a solid figure with identical and parallel end faces.
- Its other faces are parallelograms.
- If we cut a prism through a plane (flat surface) parallel to an end, the cut surface (cross-section) will be exactly the same as the end face.
- A prism is named according to the shape of the end face.
- If the end is a triangle, it is a triangular prism; if rectangular, it is a rectangular prism, and so on.

Cone

- It is a solid figure which narrows to a point (a vertex) from a circular flat base.
- It is a pyramid with a circular base.

Frustum or frustrum

- When a cone or pyramid is cut along a plane parallel to the base, the bottom part is referred to as a frustum (or frustrum).

Cylinder

- This is a hollow or solid figure with straight sides and circular ends.
- It comprises of two identical circular end faces and a curved surface.
- A cylinder is a prism with a circular cross-section.

Sphere

This is a solid figure that is entirely round, such as a ball. Half of a sphere is

called a hemisphere.

The circle

The area of a circle = π

Example

Find the area of the circle that follows

Solution

Area of the circle = π

Radius = $\frac{1}{2}$ of diameter = $\frac{1}{2} \times 14 = 7$ cm

Therefore, the area of the circle = 154 cm^2

Circumference

Circumference (C) = $\pi \times D$ D is diameter

Example

Find the diameter of the circle if the circumference is 30 cm

Solution

$C = \pi \times D$

$D = (30 \times 7 \div 2$

$= 9.55 \text{ (2 d.p.)}$

Activity

1. Find the area and diameter of the circle with the following diameters

a. 15 cm

b. 44 cm

c. 18 cm

2. Find the radius and diameter of the circle given the following circumference

a. 60 cm

b. 40 cm

Cylinder

- Area of a circular face = Area of a circle
 $= \pi r^2$
- Area of the rectangle = Circumference $\times$ height
 $= 2\pi r \times h$

∴ Surface area of a closed cylinder is given by the area of two circular faces plus the product of the circumference and the height, i.e.

- Surface area = $2\pi r^2 + 2\pi rh$
- Surface area of a closed cylinder = $2\pi r^2 + 2\pi rh$
- Surface area of an open cylinder = $\pi r^2 + 2\pi rh$
- Surface area of a hollow cylinder = $2\pi rh$
- Area of closed cylinder = $2 r(r+h)$
- Area of open cylinder (open one side) = $r(r+h)$

Example

Find the area of the closed cylinder $r = 2.8$ cm and $h = 13$ cm

Solution

$$\begin{aligned}
 &= 2 r (r+h) \\
 &= 2 \times 3.14 \times 2.8(2.8 + 13) \\
 &= 17.584 \times 15.8 \\
 &= 277.82872
 \end{aligned}$$

Note;

For open cylinder do not multiply by two, find the area of only one circle.

Example

Find the surface area of a cylinder whose radius is 7.7 cm and height 12 cm.

Solution

$$\begin{aligned}
 2 r(r+h) &= 2\pi r^2 + 2 rh \\
 S &= 2 (7.7) \times 12 + 2 (7.7) \text{ cm}^2 \\
 &= 15.4 \times 3.14 (19.7) \text{ cm}^2 \\
 &= 953.48 \text{ cm}^2
 \end{aligned}$$

Volume of a cylinder

$$\begin{aligned}
 \text{Volume of a cylinder} &= \text{Area of the circular face} \times \text{height} \\
 &= \pi r^2 h
 \end{aligned}$$

The volume of a cylinder is given by $r^2 h$

Example

Find the volume of a cylinder with height 10cm and diameter 8 cm

Solution

$$\begin{aligned}\text{Radius} &= \frac{1}{2} \text{ diameter} = 4\text{cm} \\ &= 3.14 \times 4 \times 4 \times 10 = 502.4 \text{ cm}^2\end{aligned}$$

Example

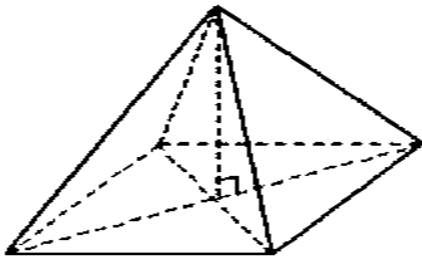
Find the volume of a cylindrical container whose diameter is 7.9 cm if its height is 7 cm. Use $\pi = 3.14$ and give the answer in 3 significant figures.

Solution

$$\begin{aligned}\text{Volume of cylinder} &= \pi r^2 h \\ &= (3.14 \times 3.952 \times 7) \text{ cm}^3 \\ &= 342.9 \text{ cm}^3 = 343 \text{ cm}^3\end{aligned}$$

Volume of a pyramid

Volume of a Pyramid The volume of a pyramid is found by:



where A is the area of the base and h is the height.

A cone is a pyramid, so the volume of a cone is given by

A square pyramid has volume

where a is the side length.

Step 1 : Determine the correct formula The volume of a pyramid is which for a square base means

$$\begin{aligned}&= \frac{1}{3} \times 2 \times 2 \times 3 \\ &= 4\text{cm}^3\end{aligned}$$

Surface area of a pyramid

- The surface area of a pyramid is the sum of the area of the slanting faces and the area of the base.

Surface area = base area + area of the four triangular faces

Example

Find the surface area of the pyramid below

Solution

Surface area = base area + area of the four triangular faces

$$= (14 \times 14) + (14 \times 14)$$

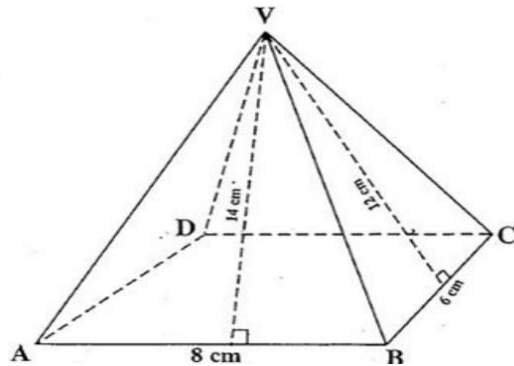
$$= 196 + 252$$

$$= 448$$

Example

Figure below shows a right pyramid with a rectangular base ABCD measuring 8 cm by 6 cm.

The heights of triangles VBC and VAB are 12 cm and 14 cm respectively.



Find the surface area of the pyramid.

Solution

Surface area = base area + area of the four triangular bases

$$= (l \times w) + 4 \left(\frac{1}{2} \text{ base} \times \text{height} \right)$$

$$\text{Area of base} = 8 \times 6 = 48\text{cm}^2$$

$$\text{Area of triangle VAB} = 8 \times 14 \div 2 = 56\text{cm}^2$$

$$\text{Area of triangle VBC} = 6 \times 12 \div 2 = 36\text{cm}^2$$

$$\text{Area of lateral face} = 56 \times 2 + 36 \times 2 = 184\text{cm}^2$$

The total surface area = $48 + 184 = 232\text{cm}^2$

Activity

1. Find the area of an open cylinder with height 15cm and radius 12 cm
2. Find the volume of a cylinder which has radius 14 cm and height 20cm
3. If the cylinder has volume 144cm^3 and diameter 16cm, find its height .
4. What is the volume of a square pyramid, 12cm high with a base 14 cm.
5. The right pyramid with a square base of 4 cm has a slanting edge of 8 cm. Find the surface area of the pyramid.

TOPIC 7 SIMULTANEOUS EQUATIONS

Linear simultaneous Equations in Two unknowns

- Many problems involve finding values of two or more unknowns.
- These are often linked via a number of linear equations.
- These are called simultaneous equations.
- There are three ways of solving simultaneous equations namely: Elimination method, substitution method and graphical method.

Elimination method

- It is used by making the coefficient of one of the variables in the two equations the same.

To solve simultaneous equations by elimination method:

- Decide which variable to eliminate.
- Make the coefficients of the variable the same in both equations.
- Eliminate the variable by addition or subtraction as is appropriate
- Solve for the remaining variable.

- Substitute the value obtained in any of the original equations to solve for the other variable.

Example

Solve the simultaneous equations

$$2x + 3y = 15,$$

$$-3y + 4x = 3.$$

Solution

Name the equations 1 and 2 as follows

$$2x + 3y = 15 \text{ (1)}$$

$$38$$

$$-3y + 4x = 3 \text{ (2)}$$

Add equations 1 and 2 in order to eliminate one variable

$$(2x + 3y) + (-3y + 4x) = 18$$

$$6x = 18$$

$$x = 3$$

Substitute $x = 3$ into (1):

$$2(3) + 3y = 15$$

$$3y = 15 - 6 \quad y = 3$$

$$x = 3, y = 3$$

Example

Using the method of elimination, solve the simultaneous equations:

$$5x + 2y = 10$$

$$4x + 3y = 1.$$

Solution

$$5x + 2y = 10 \text{ (1)}$$

$$4x + 3y = 1 \text{ (2)}$$

$$\times 3 \quad 15x + 6y = 30 \text{ (3)}$$

$$\times 2 \quad 8x + 6y = 2 \text{ (4)}$$

Subtract equation 1 by 2

$$7x = 28$$

Solve for $x = 4$

Substitute 4 for x in any one of the equations

$$4x + 3y = 1$$

$$4(4) + 3y = 1$$

$$16 + 3y = 1$$

$$3y = 1 - 16$$

$$3y = -15$$

$$y = -5$$

$$\therefore x = 4, y = -5$$

Example

The cost of two skirts and three blouses is K600. If the cost of one skirt and two blouses of the same quality K 350, find the cost of each item.

Solution

Let the cost of one skirt be x kwachas and that of one blouse be y kwachas. The cost of two skirts and three blouses is $2x + 3y$ kwachas.

The cost of one skirt and two blouses is $x + 2y$ kwachas.

$$\text{So, } 2x + 3y = 600 \dots\dots\dots (i)$$

$$x + 2y = 350 \dots\dots\dots (ii)$$

Multiplying equation (ii) by 2 to get equation (iii).

$$2x + 4y = 700 \dots\dots\dots (iii).$$

$$2x + 3y = 600 \dots\dots\dots (i)$$

Subtracting equation (i) from (iii), $y = 100$.

From equation (ii),

$$x + 2y = 350 \text{ but } y = 100$$

$$x + 200 = 350$$

$$x = 150$$

Thus the cost of one skirt is 150 kwachas and that of a blouse is 100 kwachas

Substitution method

- It is used when we make one variable the subject of an equation
- Substitute that variable into the other equation to solve for the other variable.

To solve simultaneous equations by substitution method:

- First decide which variable is easier to eliminate.
- Using the simpler of the two equations, express the variable to be eliminated in terms of the other.
- Using the other equation, substitute the equivalent for the variable to be removed.
- Solve for the remaining variable.

Example

Using the method of substitution, solve the simultaneous equations

$$5x + 2y = 10, 4x + 3y = 1.$$

Solution

$$5x + 2y = 10 \quad (1)$$

$$4x + 3y = 1 \quad (2)$$

From (1), $2y = 10 - 5x$ and $y \quad (3)$

Substitute (3) into (2):

$$8x + 3(10 - 5x) = 2$$

$$8x + 30 - 15x = 2$$

$$7x = 28$$

$$x = 4$$

Substitute $x = 4$ into (3)

$$y = -5$$

$$x = 4, y = -5$$

Example

Solve by substitution

$$2x + 3y = 600 \quad \dots\dots\dots(i)$$

$$X + 2y = 350 \quad \text{-----}(ii)$$

Taking equation (ii) alone;

Subtracting $2y$ from both sides;

$$X = 350 - 2y$$

Substituting this value of x in equation (i)

$$2(350 - 2y) + 3y = 600$$

$$700 - 4y + 3y = 600$$

$$y = 100$$

Substituting this value of y in equation (ii)

$$x + 2(100) = 350$$

$$x = 350 - 200$$

$$x = 150$$

$$x = 150 \text{ and } y = 100$$

Activity

1. Solve the following simultaneous equations: $x + 3 = 11$ $5 + 4 = 22$.
2. Solve the simultaneous equations: $y = -1$, $3 - y = 5$

Graphical solutions to simultaneous equations

- Apart from substitution method and elimination method, we can also solve simultaneous equations by using graphs.

Example

Copy and complete the table of values for $y = 5 - 2x$ and $y = x - 4$.

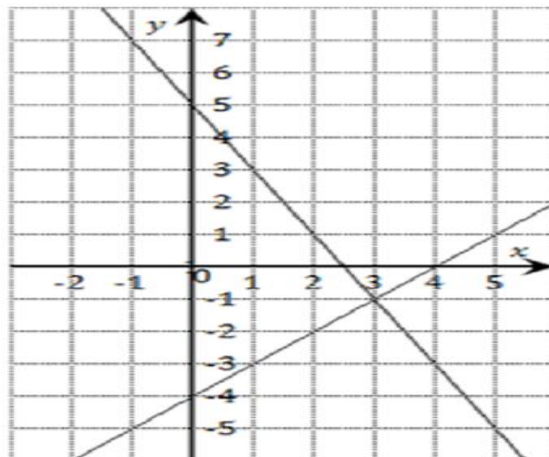
$$y = 5 - 2x$$

x	-1	2	5
y	7	1	-5

$$y = x - 4$$

x	-1	2	5
y	-5	-2	1

- b. Using a scale of 2 cm to represent 1 unit on the x -axis and 2 cm to represent 1 unit on the y -axis, draw on the same axes, the graphs of $y = 5 - 2x$ and $y = x - 4$.



- c. Use the graph to solve the simultaneous equations: $y = 5 - 2x$ and $y = x - 4$. (2 marks)

On the graph, the line $y = 5 - 2x$ and $y = x - 4$ meet at the point $(3, -1)$.

$\therefore x = 3$ and $y = -1$

Forming and solving simultaneous equations

Example

Two years ago, a man was seven times as old as his son. In three years time, he will be only four times as old as his son. Find their present ages.

Solution

If the present age of the man = m and the present age of the son = s years,

Then two years ago, the man's age was $(m - 2)$ years, and the son's age was $(s - 2)$ years.

$$\therefore m - 2 = 7(s - 2) \dots\dots\dots (1)$$

In three years time, the man's age = $(m + 3)$

In three years time, the son's age = $(s + 3)$

$$\therefore m + 3 = 4(s + 3) \dots\dots\dots (2)$$

Equations (1) and (2) can be written simply as

$$m = 7s - 12 \dots\dots\dots (3)$$

$$m = 4s + 9 \dots\dots\dots (4)$$

Since the LHS are equal, the RHS are also equal.

$$\therefore 7s - 12 = 4s + 9$$

$$3s = 21$$

$$s = 7$$

Substituting 7 for s in (3):

$$m = 7 \times 7 - 12$$

$$= 49 - 12$$

$$= 37$$

The present age of the man is 37 years.

The present age of the son is 7 years.

Example

A two digit number is such that its value equals four times the sum of its digits. If 27 is added to the number, the result is equal to the value of the number obtained when the digits are interchanged. What is the number?

Solution

Let the tens digit be x .

Let the ones digit be y .

$\therefore$ the value of the number is $10x + y$ and the sum of the digits is $x + y$.

$$10x + y = 4(x + y)$$

$$10x + y = 4x + 4y$$

$$6x = 3y$$

$$2x = y$$

$$2x - y = 0 \dots\dots\dots (1)$$

The value of the number formed by interchanging the digits is $10y + x$.

$$\therefore 10x + y + 27 = 10y + x.$$

$$9x - 9y + 27 = 0$$

$$9x - 9y = -27$$

$$x - y = -3 \dots\dots\dots (2)$$

Subtract (2) from (1):

Substitute $x = 3$ in (1):

$$(2 \times 3) - y = 0$$

$$6 - y = 0$$

$$\therefore y = 6$$

The original number is 36.

TOPIC 8 RATES, RATIO, PROPORTION AND PERCENTAGES

Rate

- A rate is a measure of quantity, and comparing one quantity with another of different kind

Example

What is the rate of consumption per day if twelve bags of beans are consumed in 120 days?

Solution.

Rate of consumption = number of bags/number of days
=1/10 bags per day

Example

A laborer's wage is k240 per eight hours working day. What is the rate of payment per hour?

Solution

Rate = amount of money paid/number of hours

$$= 240$$

=k30 per hour

Ratio

- A ratio is a way of comparing two similar quantities.
- In stating a ratio, the units must be the same.
- A ratio is expressed in its simplest form in the same way as a fraction.

Proportion

- A proportion is a comparison of two or more ratios.
- If example, a, b and c are three numbers such that a: b: c=2:3:5, then a, b, c are said to be proportional to 2, 3, 5 and the relationship should be interpreted to mean $a/2 = b/3 = c/5$.

Similarly, we can say that a: b =2:3, b: c=3:5 a: c=2:5

Example 3

If a: b = 3: 4 and b: c = 5: 7 find a: c

Solution

$$a: b = 3 : 4 \dots\dots\dots(i)$$

$$b: c = 5 : 7 \dots\dots\dots(ii)$$

Consider the right hand side;

Multiply (i) by 5 and (ii) by 4 to get,

$$a: b = 15: 20 \text{ and } b: c = 20: 28$$

Thus, a: b: c = 15: 20: 28 and a: c=15: 28

Increase and decrease in a given ratio

- To increase or decrease a quantity in a given ratio, we express the ratio as a fraction and multiply it by the quantity.

Example

Increase 20 in the ratio 4 : 5

Solution

New value $= \frac{5}{4} \times 20$

$$= 5 \times 5$$

$$= 25$$

Example

Decrease 45 in the ratio 7:9

Solution

New value $= \frac{7}{9} \times 45$

$$= 7 \times 5$$

$$= 35$$

Example

The price of a pen is adjusted in the ratio 6:5. If the original price was k50. What is the new price?

Solution

New price: old price = 6:5

New price / old price = $\frac{6}{5}$

New price = $\frac{6}{5} \times 50 = \text{k}60$

Note:

When a ratio expresses a change in a quantity an increase or decrease, it is usually put in the form of new value: old value

Comparing ratios

In order to compare ratios, they have to be expressed as fractions first, ie., $a:b = \frac{a}{b}$ the resultant fraction can then be compared.

Example

Which ratio is greater, 2: 3 or 4: 5?

Solution

$$2:3 = \frac{2}{3},$$

$$4:5 = \frac{4}{5}$$

$$\frac{2}{3} = \frac{10}{15}, \frac{4}{5} = \frac{12}{15} = \frac{4}{5} > \frac{2}{3}$$

Thus, $4: 5 > 2: 3$

Distributing a quantity in a given ratio

If a quantity is to be divided in the ratio $a : b : c$, the fraction of the quantity represented by:

(i) A will be

(ii) B will be

(iii) C will be

Example

A 72-hactare farm is to be shared among three sons in the ratio 2:3:4. What will be the sizes in hectares of the three shares?

Solution

Total number of parts is $2+3+4=9$

The she shares are: $\frac{2}{9} \times 72\text{ha} = 16\text{ha}$

$\frac{3}{9} \times 72\text{ha} = 24\text{ha}$

$\frac{4}{9} \times 72\text{ha} = 36\text{ha}$

Direct and inverse proportion

Direct proportion

- If two quantities are such that when the one increases (decreases) in particular ratio, the other one also increases (decreases) in the ratio.

Example

A car travels 40km on 5 litres of petrol. How far does it travel on 12 litres of petrol?

Solution

Pet Distance = $40 \times \frac{12}{5}$ km

= 96km

Example

A train takes 3 hours to travel between two stations at an average speed of 40km per hour. A t what average speed would it need to travel to cover the same distance in 2 hours?

Solution

Time is decreased in the ratio 2:3

Speed must be increased in the ratio 3:2

Average speed is $40 \times \frac{3}{2} \text{ km} = 60 \text{ km/h}$

Example

Ten men working six hours a day take 12 days to complete a job. How long will it take eight men working 12 hours a day to complete the same job?

Solution

Number of men decreases in the ratio 8:10

Therefore, the number of days taken increases in the ratio 10:8.

Number of hours increased in the ratio 12:6.

Therefore, number of days decreases in the ratio 6:12.

Number of days taken $= 12 \times x$

$= 7 \frac{1}{2}$ days

Percentages

- A percentage (%) is a fraction whose denominator is 100.
- For example, 27% means $\frac{27}{100}$.

Converting fractions and decimals into percentages

- To write a decimal or fraction as a percentage (%) multiply by 100 .

Example

$0.125 = 0.125 \times 100 = 12.5\%$

Example

Change $\frac{2}{5}$ into percentage.

Solution

$= \frac{2}{5} \times 100\%$

$= 40\%$

Percentage increase and decrease

A quantity can be expressed as a percentage of another by first writing it as a fraction of the given quantity.

Example

A man earning K4 800 per month was given a 25% pay rise. What was his new salary?

Solution

$$\begin{aligned}\text{New salary} &= 25/100 \times 4800 + 4\,800 \\ &= 1\,200 + 4\,800 \\ &= \text{K } 6\,000\end{aligned}$$

Example

A dress which was costing K1 200 now goes for K960. What is the percentage decrease?

Solution

$$\begin{aligned}\text{Decrease in cost} &= 1\,200 - 960 = 240 \\ \text{Percentage decrease} &= 240/1\,200 \times 100 \\ &= 20\%\end{aligned}$$

Example

The ratio of John's earnings to muse's earnings is 5:3. If John's earnings increase by 12%, his new figure becomes k5 600. Find the corresponding percentage change in Muse's earnings if the sum of their new earnings is K 9 600

Solution

$$\begin{aligned}\text{John's earnings before the increase} &= 100/112 \times 5600 = \text{K } 5\,000 \\ \text{John's earnings/Muse's earnings} &= 5/3 \\ \text{Musa's earnings before the increase} &= 3/5 \times 5000 \\ &= \text{K } 3\,000 \\ \text{Musa's new earnings} &= 9\,600 - 5\,600 \\ &= \text{K } 4\,000 \\ \text{Musa's change in earnings} &= 4\,000 - 1\,000 \\ &= \text{K } 3\,000 \\ \text{Percentage change in muse's earnings} &= 1000/3000 \times 100 \\ &= 33\%\end{aligned}$$

TOPIC 9 DENSITY AND MIXTURES

DENSITY

- Mass is the quantity of matter in a substance.
- The mass of an object is found by comparing it with a standard mass using a beam balance.
- The gram or gramme (abbreviated as g) is the most common unit of mass.
- One gram is the mass of 1 c (1 ml) of water at a temperature of 4°C.
- The SI unit of mass is the kilogram or kilogramme (kg).
- The density of a substance is the mass of a unit volume of that substance.
- It is given by the formula:

$$\text{Density} = \frac{\text{Mass}}{\text{volume}}$$

- If mass is in g and volume is in cm³, then density is measured in g/cm³.
- If the mass is in kg and volume is in m³, then density is measured in kg/m³.
- To convert density from g/cm³ to kg/m³ we multiply by 1 000, while from kg/m³ to g/cm³ we divide by 1 000.

Example

A rectangular tank of volume 1800cm³ is filled with fuel. Given that the density of fuel is 0.7g/cm³, calculate the mass of fuel in the tank.

Solution

$$\begin{aligned}\text{Mass} &= \text{volume} \times \text{density} \\ &= 1800\text{cm}^3 \times 0.7\text{g/cm}^3 \\ &= 1260\text{g}\end{aligned}$$

Relative density

- It is the density of a substance is compared to the density of a known substance.
- Water being the most common and with a density of 1 g/cm³ is used as the reference substance.

$$\text{Relative density} = (\text{Mass of substance}) \div (\text{Volume of substance})$$

Example

A mathematical instrument box has a mass of 273 g. If its relative density is 2.1, calculate the volume of the mathematical instrument box.

Solution

$$\text{Relative density} = \text{Mass of substance} / \text{Volume of substance}$$

Let the Volume of mathematical instrument box = x

$$\begin{aligned}2.1 &= \frac{273}{x} \\ x &= \frac{273}{2.1} \\ &= 130\end{aligned}$$

Example

Density of mixtures

- Two or more substances are mixed in a given proportion.
- The density of the mixture is found from the density of the substances used to make it.

Example

A bottle weighs 20 g when empty, 55 g when full of petrol and 70 g when full of water. Calculate the relative density of petrol.

Solution

The weight of petrol = $55 - 20 = 35\text{g}$

The weight of water = $70 - 20 = 50\text{g}$

Relative density = $35\text{g}/50\text{g} = 0.7$

Mixtures

- A mixture is a combination of two or more substances.
- Business people mix different commodities before selling them in order to minimise on costs while maximising on profits. For example, in the market you may find one type of bean mixed with another.

Example

In what ratio must white beans costing K146 per kg be mixed with red beans costing K170 per kg in order for the mixture to cost K152 per kg?

Solution

Let quantity of white beans be x and that of red beans be y

$$(146 \times x) + (170 \times y) = (x + y)152 \Rightarrow 146x + 170y = 152x + 152y$$

$$170y - 152y = 152x - 146x \Rightarrow \frac{18y}{6} = \frac{6x}{6} \Rightarrow \frac{y}{x} = \frac{3}{1}$$

The ratio of white beans to red is 3: 1

Activity

1. A cylinder of internal diameter 14 cm and height 10 cm is filled with a fluid of mass 3080g. Calculate the density of the fluid. (Take $\pi = 22$)
2. A bottle weighs 77 g when empty, 101 g when full of water and 107 g when full of glycerine. Find the relative density of glycerine.
3. A rectangular tank of volume 1800c is filled with fuel. Given that the density of fuel is 0.7g/c, calculate the mass of fuel in the tank.
4. A mathematical instrument box has a mass of 273 g. If its relative density is 2.1, calculate

the volume of the mathematical instrument box.

TOPIC 10 INDICES AND LOGARITHMS

- The power to which a number is raised is called index or indices in plural. e.g. 10^2 , 2 is called the index and 10 is the base.

Laws of indices

- For the laws to be applied the base must be the same.

Rule 1: Any number, except zero whose index is 0 is always equal to 1

Example

$$5^0 = 1$$

Rule 2 : To multiply an expression with the same base, copy the base and add the indices.

$$x \times x = x^2$$

Example

$$2^3 \times 2^2 = 2^{(3+2)} = 2^5 = 32$$

Rule 3 : To divide an expression with the same base, copy the base and subtract the powers.

$$x^a \div x^b = x^{a-b}$$

Example

$$2^6 \div 2^2 = 2^{(6-2)} = 2^4 = 16$$

Rule 4 : To raise an expression to the nth index, copy the base and multiply the indices

$$(X^a)^b = x^{ab}$$

Example

$$(2^2)^3 = 2^{(2 \times 3)} = 2^6 = 32$$

Rule 5: When dealing with a negative power, change the power to positive by changing it into a fraction with 1 as the numerator.

$$x^{-b} = \frac{1}{x^b}$$

Example

$$2^{-3} = \frac{1}{2^3} = \frac{1}{8}$$

Rule 6: Fractional indices are written in fraction form. If a is called the root of b it is written as

$$x^{\frac{a}{b}} = \sqrt[b]{x^a}$$

Example

$$8^{\frac{2}{3}} = \sqrt[3]{8^2} = 2^2 = 4$$

Logarithm

Logarithmic function definition

- The logarithmic function is defined as an inverse function to exponentiation.
- The logarithmic function is stated as follows

For $x, a > 0$, and $a \neq 1$,

$$y = \log_a x, \text{ if } x = a^y$$

Then the logarithmic function is written as:

$$f(x) = \log_a x$$

- The most 2 common bases used are base e and base 10.
- The log function with base 10 is called the common logarithmic function.
- It is denoted by $\log_{10}$ or simply $\log$. $f(x) = \log_{10}$
- The log function to the base e is called the natural logarithmic function and it is denoted by $\log_e$.

$$f(x) = \log_e x.$$

Finding logarithm of a number

Using calculators to find logarithms and antilogarithms.

Example

Find the logarithm of 0.00063

Solution

(Find the logarithm of 6.3)

$$= 0.7993$$

$$= - 4.7993$$

Example

Find the logarithm of: 379

Solution

$$379 = 3.79 \times 10^2$$

$$\text{Log } 3.79 = 0.5786$$

Antilogarithms

- Finding antilogarithm is the reverse of finding the logarithms of a number. For example, the logarithm of 1000 to base 10 is 3.
- So, the antilogarithm of 3 is 1000. In algebraic notation, if

$$\text{Log } x = y \text{ then antilog of } y = x.$$

Example

Find the antilogarithm of 0.3031

Solution

Let the number be x

(Find the antilog, press shift and log then key in the number)

$$= 0.5184$$

Example

Use logarithm tables to evaluate: $456 \div 398 \times 271$

Number Standard form logarithm

$$456 = 4.56 \times 10^2 \quad \log 4.56 = 0.6590$$

$$398 = 3.98 \times 10^2 \quad \log 3.98 = 0.5999 \text{ add}$$

$$0.6590 + 0.5999$$

$$= 1.2589$$

$$271 = 2.71 \times 10^2 \quad \log 2.71 = 0.4330 \text{ subtract } 1.2589 - 0.4330$$

$$= 0.8259$$

Next Find antilog of 0.8259

To find the exact number find the antilog of 0.8259 by letting the characteristic part to be the power of ten then finding the antilog of 0.8259. The mantissa (which is 8259) in this case will determine where to place a decimal point. In this case the answer must be between 100 and 1000 = 669.7

When numbers are multiplying, we add the logs

When numbers are dividing, we subtract the logs

When numbers has a power e.g. 3456 , we find the log of $345 = 0.5378$ and then multiply by 6 = 3.2269 and then find its antilog = 1.686×10^3

When a number has a root e.g. $\sqrt[3]{234}$, we find log of $234 = 0.3692$ and then we divide by 3 = 0.1231 and then find the antilog of 0.1231.

Example

Evaluate using log tables:

Number Standard form Logarithm

$$3456 = 3.456 \times 10^3 \quad \log 3.456 = 0.5390$$

$$235 = 2.35 \times 10^2 \quad \log 2.35 = 0.3711 \times 2 = 0.7422$$

$$\text{Subtract } 0.5390 - 0.7422 = \bar{0.2032}$$

Antilog of $\bar{0.2032} = 0.0625 = 6.25 \times 10^{-2}$ the ($\bar{4}$ means shifting the decimal point to the left 4 times).

Activity

Evaluate using log tables

- a. $415.2 \times 0.0761 \div 135$ b. $\log 0.02343$ c. $\log 2567$

TOPIC 11 SIMILARITY I

Two solids are similar if

1. the ratio of the lengths of their corresponding sides is constant
2. the corresponding angles are equal

Example

A jewel box, of length 30 cm, is similar to a matchbox. If the matchbox is 5 cm long, 3.5 cm wide and 1.5 cm high, find the breadth and height of the jewel box.

Solution

The ratio of the lengths of the jewel box and the matchbox is

$$\frac{\text{Length of jewel box}}{\text{Length of match box}} = \frac{30 \text{ cm}}{5 \text{ cm}} = 6$$

This means that each edge of the jewel box is 6 times the length of the corresponding edge of the matchbox = $6 \times 3.5 \text{ cm} = 21 \text{ cm}$,

height of jewel box = $6 \times 1.5 \text{ cm}$

$$= 9 \text{ cm}.$$

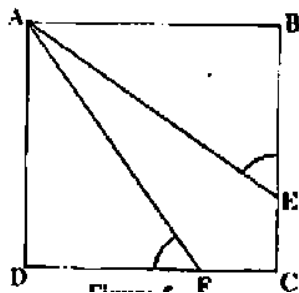
Similar triangles

Two triangles are similar if either

1. the ratio of corresponding sides is constant
2. the corresponding angles are equal.

Example

In Figure below, ABCD is a square.



If angle BEA = Angle DFA, show that triangle BEA is similar to triangle DFA.

Solution

In triangle BEA and triangle DFA,

$$\angle AFD = \angle BEA \text{ (given)}$$

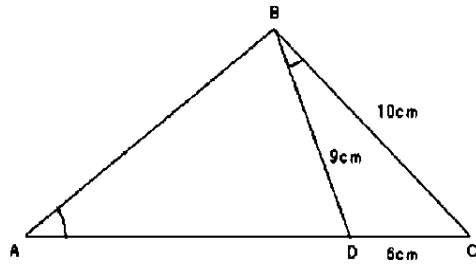
$\angle ADF = \angle ABE = 90^\circ$ (ABCD is a square)

$\angle DAF = \angle BAE$ (remaining angles of $\triangle BEA$ and $\triangle BCD$)

$\therefore$ triangle BEA is similar to triangle DFA

Example

Figure below is triangle ABC similar to triangle BDC. Angle BAC = angle DBC, BD = 9 cm, BC = 10 cm and DC = 6 cm. Calculate the length of AB.



Solution.

$$\frac{AB}{BD} = \frac{BC}{DC}$$

$$\frac{AB}{9} = \frac{10\text{cm}}{6\text{cm}}$$

$$AB = \frac{10}{6} \times 9 = 15\text{cm}$$

TOPIC 12 INEQUALITIES

- Statements connected by these symbols are called inequalities

Representing linear inequalities on a number line

Simple statements

- Simple statements represent only one condition on a number line as follows:
 - a. $X = 3$ represents specific point which is number 3, while $x > 3$ does not represents all numbers to the right of 3 meaning all the numbers greater than 3 as illustrated above.
 - b. $X < 3$ represents all numbers to left of 3 meaning all the numbers less than 3. The empty circle means that 3 is not included in the list of numbers to greater or less than 3. The expression means that means that 3 is included in the list and the circle is shaded to show that 3 is included.

Compound statement

- A compound statement is a two simple inequalities joined by “and” or “or.” Here are two examples.

Solution to simple inequalities

Example

Solve the inequality $x - 1 > 2$

Solution

Adding 1 to both sides gives ;

$$X - 1 + 1 > 2 + 1$$

Therefore, $x > 3$

Note;

In any inequality you may add or subtract the same number from both sides.

Example

Solve the inequality $x + 3 < 8$

Solution

Subtracting three from both sides gives

$$X + 3 - 3 < 8 - 3$$

$$X < 5$$

Example

Solve the inequality $2x + 3 > 2$

Solution

Subtracting three from both sides gives

$$2x + 3 - 3 > 2 - 3$$

$$2x > -1$$

Divide both sides by 2 gives

$$x >$$

Multiplication and Division by a negative number

- Multiplying or dividing both sides of an inequality by positive number leaves the inequality sign unchanged
- Multiplying or dividing both sides of an inequality by negative number reverses the sense of the inequality sign.

Example

Solve the inequality $1 - 3x < 4$

Solution

$$-3x - 1 < 4 -$$

$$-3x < 3 \quad x > -1 \text{ (divide by } -3 \text{ to both sides)}$$

Note that the sign is reversed $x > -1$

Example

Solve the following

$$3x - 1 > -4$$

Solution

Solving the first inequality

$$3x - 1 > -4$$

$$3x > -3$$

$$x > -1$$

Example

Solve $4q + 3 < 2(q + 3)$ and represent the solution on a number line.

Solution

Step 1 : Expand all brackets

$$4q + 3 < 2(q + 3)$$

$$4q + 3 < 2q + 6$$

Step 2 : Move all constants to the RHS and all unknowns to the LHS

$$4q + 3 < 2q + 6$$

$$4q - 2q < 6 - 3$$

$$2q < 3$$

Step 3 : Solve inequality

$$2q < 3 \text{ Divide both sides by 2}$$

$$q < 3/2 \text{ or } q < 1.5$$

Formulating inequalities from word problems**Example**

A student spent $(8y + 40)$ kwacha on biscuits and $2y$ kwacha on sweets. If the total amount of money spent was less than K80, formulate an inequality from this information.

Solution

$$(8y + 40) + 2y < 80$$

$$10y + 40 < 80$$

$$10y < 40$$

$$y < 4$$

TOPIC 13 DISTANCE TIME-GRAPHS

Distance

- The length from one point to another is known as the distance.
- It is measured in metres (m) .
- For long distances, kilometre (km) is use while the shortest distance is in metre (m).
- Other shorter distances may be measured in centimetres (cm) or in millimetres (mm).

Speed

Speed is amount of distance covered by an object per unit time.

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

If the distance is in kilometres and the time is in hours, then the speed is given in kilometres per hour (km/h or km/h or kph). If the distance is in metres and the time in seconds, then speed is given in metres per second (m/s or m/s).

Example

A train leaves town A and travels towards town B at 48 km/h. At the same time, another train leaves town B and travels towards A at a speed of 52 km/h. If the two towns are 500 km apart, find

- how far apart the trains are after travelling for 45 minutes.
- how far, from town A, they will be when they meet.
- how long they will take to meet.

Solution

Distance = Time $\times$ Speed

Train from town A will have travelled a distance and train from town B will have travelled a distance

$$\begin{aligned}\therefore \text{the distance apart} &= 500 - (36 + 39) \text{ km} \\ &= 425 \text{ km}\end{aligned}$$

(b) The trains start at the same time. So they have travelled the same length of time by the time they meet.

$$\Rightarrow 24\,000 - 48x = 52x$$

$$\Rightarrow 24\,000 = 52x + 48x$$

$$\Rightarrow 24\,000 = 100x$$

$$\therefore x = 240 \text{ km}$$

i.e. the trains meet 240 km away from town A.

Distance time graph

On a distance–time graph:

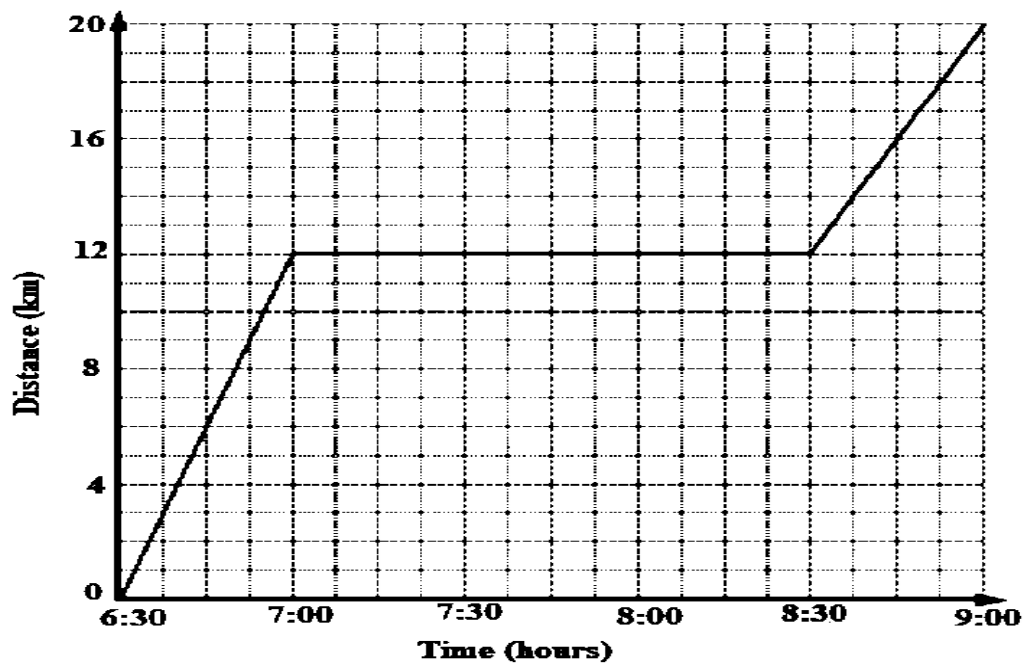
- The vertical axis represents the distance from the starting point.
- The horizontal axis represents the time taken.
- A horizontal line represents no movement.
- The gradient of the slope gives the speed (a straight line implies a constant speed).
- A positive gradient represents the outbound journey.
- A negative gradient represents the return journey.

Example

Chikondi left point A at 6:30 am for point C, which is 20 km away. At 7:00 am he stopped at point B for hours having covered a distance of 12 km. He then continued at a speed of 16 km/h.

- Using a scale of 2 cm to represent 4 km on the vertical axis and 2 cm to represent 30 minutes on the horizontal axis, draw the distance-time graph for the cyclist.
- Using your graph, find the time taken for the cyclist to reach his destination.

Solution

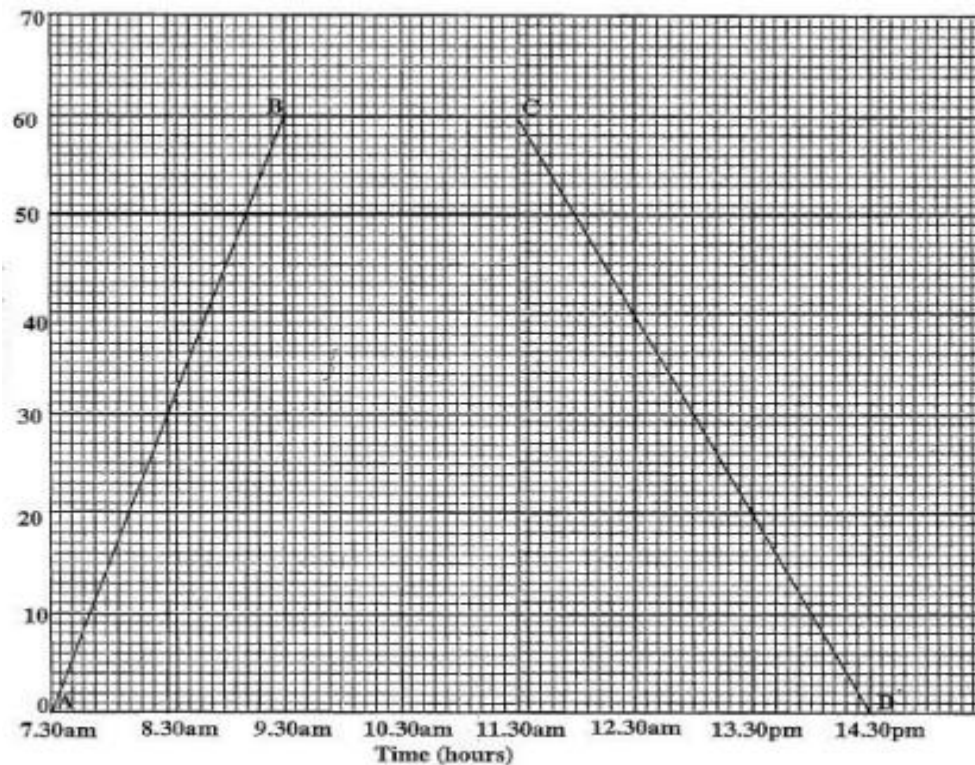


b. Using your graph, find the time taken for the cyclist to reach his destination.

= 2.5 hours

Example

Figure below shows a graph ABCD of Mr. Mandebvu's journey.



- (i) How far is B from A?
- (ii) Calculate the average speed of Mr. Mandebvu's journey from C to D.

Solution

(i) 60 km

$$\begin{aligned}
 \text{(ii) speed} &= \frac{\text{distance}}{\text{time}} = \frac{\text{Distance from C to D}}{\text{Time 11.30am to 14.30pm}} \\
 &= \frac{60\text{km}}{3 \text{ hours}} \\
 &= 20\text{km/h}
 \end{aligned}$$

Activity

1. Dennis travels south on a car from Kasungu, while Franz travels north on the same road from Mzimba. This distance–time graph (showing the distance from Mzimba) shows the journeys of both travellers.

a What is Ingar's speed in kilometres per hour?

b What is Franz's speed in kilometres per hour?

c. At what time does Ingar reach Hannover? d What is the distance between Ingar and Franz at 10:30?

e At what time do Ingar and Franz pass each other ?

2. A passenger notices that she had forgotten her bag in a bus 12 minutes after the bus had left. To catch up with the bus she immediately took a taxi which traveled at 95 km/hr. The bus maintained an average speed of 75 km/ hr. Determine :

a. The distance covered by the bus in 12 minutes

b. The distance covered by the taxi to catch up with the bus

TOPIC 14 PROBABILITY

- Is the likelihood of an occurrence of an event or the numerical measure of chance.

Experimental probability (single event experiments)

- This is where probability is determined by experience or experiment.
- It is done or observed is the experiment.
- Each toss is called a trial and the result of a trial is the outcome.

The experimental probability of a result is given by:

Probability =

Eg Tossing a die.

Example

A boy had a fair die with faces marked 1 to 6 . He threw this die once and the results of his experiment is shown below.

Face 1 2 3 4 5 6

All possible outcomes are: 1,2,3,4,5,6

What is the experimental probability of getting?

a. 1 **b.** 6 **c.** an odd and an even number **d.** a prime number **e.** a multiple of 3

Solution

a. $P(1) = 1/6$

b. $P(6) = 1/6$

c. $P(\text{odd number}) = 1/6 + 1/6 + 1/6 = 3/6 = 1/2$ (there are three odd numbers: 1,3 and 5)

d. $P(\text{odd + even number}) = 6/6 = 1$ (3 odd numbers and 3 even numbers)

$P(\text{Prime number}) = 3/6 = 1/2$ (3 prime numbers)

e. $P(\text{multiple of 3}) = 2/6 = 1/3$ (2 multiples of 3)

Tossing a coin

A coin has two sides : head and tail. These two sides have fair chance when the coin is tossed.

Example : If a coin is tossed, find

- i.) The probability of getting a head or a tail
- ii.) Probability of getting a head and a tail
- iii.) Probability of getting a head

Solution

i. $P(\text{head}) = \frac{1}{2}$

ii. $P(\text{head}) + P(\text{tail}) = \frac{1}{2} + \frac{1}{2} = 1$

Note; If OR is used then we add and when AND is used we multiply

iii. $P(\text{head and tail}) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

Throwing cards

Theoretical probability

- This can be calculated without necessarily using any past experience or doing any experiment.
- The probability of an event happening = $\frac{\text{number of favourable outcomes}}{\text{total number of possible outcomes}}$.

Example

A basket contains 5 red balls, 4 green balls and 3 blue balls. If a ball is picked at random from the basket, find:

- The probability of picking a blue ball
- The probability of not picking a red ball

Solution

Total number of balls is 12

The number of blue balls is 3

a. therefore, $P(\text{a blue ball}) = \frac{3}{12}$

b. The number of balls which are not red is 7.

Therefore $P(\text{not a red ball}) = \frac{7}{12}$

Example

A bag contains 6 black balls and some brown ones. If a ball is picked at random the probability that it is black is 0.25. Find the number of brown balls.

Solution

Let the number of balls be x

Then the probability that a black ball is picked at random is $\frac{6}{x}$

Therefore $\frac{6}{x} = 0.25x$

$$= 24$$

The total number of black is 24

Then the number of brown balls is $24 - 6 = 18$

Activity

1. In a car park with 300 cars, there are 190 Opals. What is the probability that the first car to leave the car park is:

- a.an Opal
- b.not an Opal

2. Tamara has 18 loose socks in a drawer. Eight of these are orange and two are pink. Calculate the probability that the first sock taken out at random is:

- a.Orange
- b.not orange pink
- c.not pink
- d.orange or pink
- c.not orange or pink

3.A plate contains 9 shortbread cookies, 4 ginger biscuits, 11 chocolate chip cookies and 18 Jambo's. If a biscuit is selected at random, what is the probability that:

- a. it is either a ginger biscuit OR a Jambo?
- b. it is NOT a shortbread cookie.

3. A jar has purple, blue and black sweets in it. The probability that a sweet, chosen at random, will be purple is $\frac{1}{7}$ and the probability that it will be black is $\frac{3}{5}$. If I choose a sweet at random what is the probability that it will be:

- a.purple or blue
- b. Black
- c. purple
- d.If there are 70 sweets in the jar how many purple ones are there?

TOPIC 15 GEOMETRIC CONSTRUCTIONS II

Inscribed circle

Steps to follow

- Construct any triangle ABC.
- Construct the bisectors of the three angles
- The bisectors will meet at point I
- Construct a perpendicular from O to meet one of the sides at M
- With the centre I and radius IM draw a circle

The circle will touch the three sides of the triangle ABC. Such a circle is called an inscribed circle or in circle.

The centre of an inscribed circle is called the incentre.

Example

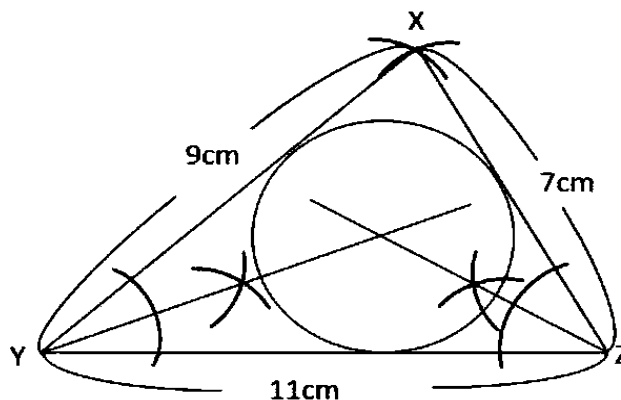
a. Using a ruler and a pair of compasses only, Construct in the same diagram:

(i) a triangle XYZ in which $XY = 9\text{ cm}$, $YZ = 11\text{ cm}$ and $ZX = 7\text{ cm}$.

(ii) the inscribed circle of the triangle XYZ.

b. Measure and state the length of the radius of the circle.

Solution



Circumscribed circle

Steps to follow

- Construct any triangle ABC.
- Construct perpendicular bisectors of AB, BC, and AC to meet at point O.

- c. With O as the centre and using OB as radius, draw a circle
- d. The circle will pass through the vertices A , B and C as shown in the figure below

Example

Using a ruler and a pair of compasses only, construct in the same diagram:

- (i) a triangle PQR in which $PQ = QR = PR = 5\text{cm}$.
- (ii) construct the circumscribed circle of the triangle PQR.

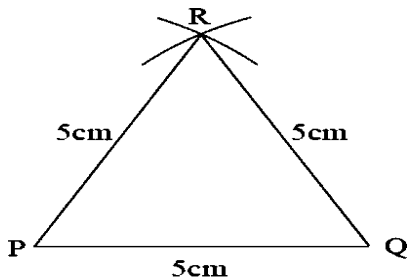
Solution

Step1. Using a ruler, draw a line of 5cm. Label the ends, P and Q.

Step2. Set the compasses to the second length, 5cm. Place the point of the compasses at P and draw an arc.

Step3. Set the compasses to the third length, 5cm. Place the point of the compasses at Q and draw another arc. Label the intersection, R.

Step4. Join P and Q to the intersection R.

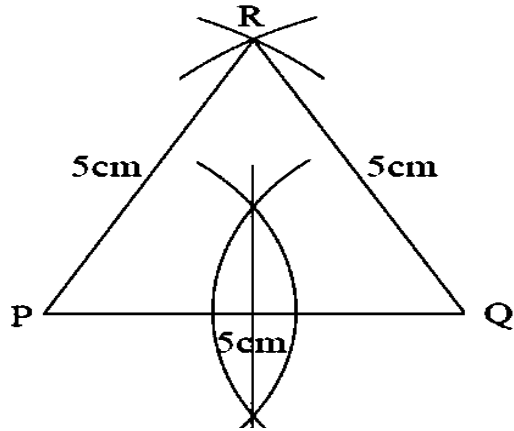


a. (ii)

Step1. Set the compasses to slightly more than half the length of the side PQ. Place the point at P and draw an arc.

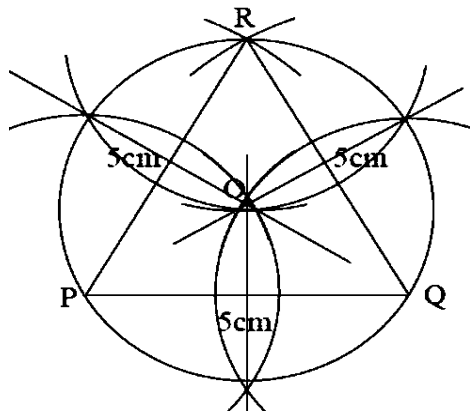
Step2. Without changing the setting of the compasses, place the point at Q and draw another arc.

Step3. Join the intersections of the arcs to find the perpendicular bisector.



Step 4. Do “Step1 to Step3” also about sides PR and QR. The intersection of the three perpendicular bisectors is the centre (O) of the circumscribed circle.

Step 5. Set the compasses to the length of OP. Place the point of the compasses at O and draw the circle.



Activity

1. Using a ruler and a pair of compasses only, Construct in the same diagram:

- (i) a triangle XYZ in which $XY = 9$ cm, $YZ = 11$ cm and $ZX = 7$ cm.
- (ii) the inscribed circle of the triangle XYZ.

b. Measure and state the length of the radius of the circle.

2. Using a ruler and a pair of compasses only, construct in the same diagram:

- (i) a triangle PQR in which $PQ = QR = PR = 5$ cm.
- (ii) construct the circumscribed circle of the triangle PQR

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